



MECH MAINT- TURBINE Department
SIMHADRI

Ref: SMTTP/MECH MAINT-/2021-22/MM/TMD/788189

Date:

11/02/2022(17:34:31)

Subject: Revision of LMI "Operation and Maintenance Practices of HVAC Systems" LMI NO. LMI/OGN/OPS/MECH/044- Reg

The LMI: "Operation and Maintenance Practices of HVAC Systems" with LMI NO. LMI/OGN/OPS/MECH/044 is due for review in the month of Jan'2022. The LMI is reviewed and scope of the LMI included at S.NO: 3 as per the last audit findings.

CC-OS had reviewed the document in Sep'2020 and no changes were envisaged.

Put up for Approval of Competent authority.

Encl:

- Earlier approved LMI.
- New LMI

Submitted by:

रामबाबू वानापल्ली - उप महाप्रबंधक

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On: 11/02/2022(17:34:31) **Ref:** SMTTP/MECH MAINT-/2021-22/MM/TMD/788189

Submitted for approval please.

Submitted by:

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On:11/02/2022(17:59:46) **Ref:**SMTTP/MECH MAINT-/2021-22/MM/TMD/788189

Please review clause no. 6.0 (O&M practices for HVAC systems) as discussed.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:12/02/2022(15:56:00) **Ref:**SMTTP/MECH MAINT-/2021-22/MM/TMD/788189

Pl confirm LMI slno 6.6 daily walk down check list data

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On:12/02/2022(16:01:44) **Ref:**SMTTPP/MECH MAINT-/2021-22/MM/TMD/788189

Records are being maintained against clause 6.0.

Submitted by:

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Clause 6.0 reviewed.

May pl process

Submitted by:

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On:12/02/2022(17:11:20) **Ref:**SMTTPP/MECH MAINT-/2021-22/MM/TMD/788189

LMI reviewed and found OK. Approval may please be accorded by Competent Authority.

Submitted by:

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On:12/02/2022(17:21:07) **Ref:**SMTTPP/MECH MAINT-/2021-22/MM/TMD/788189

May pl see before approval.

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On:12/02/2022(18:32:25) **Ref:**SMTTPP/MECH MAINT-/2021-22/MM/TMD/788189

Chkd and found ok.

Put up for approval of competent authority

Submitted by:

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On:14/02/2022(09:11:14) **Ref:**SMTTPP/MECH MAINT-/2021-22/MM/TMD/788189

Submitted by:

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On:14/02/2022(09:28:48) **Ref:**SMTPP/MECH MAINT-/2021-22/MM/TMD/788189

Approved & Submitted by:

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On:14/02/2022(16:03:28) **Ref:**SMTPP/MECH MAINT-/2021-22/MM/TMD/788189



SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/044

ISSUE NO.1; REVISION NO. 02; DATE: FEB'2022

**GUIDELINES FOR OPERATION AND MAINTENANCE
OF HVAC SYSTEMS**

Approved for

Implementation _____
HOP (SIMHADRI)

Date: _____



SIMHADRI

LMI APPROVAL DOCUMENT

TITLE OF LMI	Operation and Maintenance practices of HVAC systems.
DOCUMENT CODE	LMI/OGN/OPS/MECH/044
REVISION / ISSUE NO & DATE	ISSUE NO.1, REV.02, DATE: FEB'2022
ORIGINAL / REF. DOCUMENT	COS-ISO-00-OGN/OPS/MECH/044 Rev.No:O
4.PREPREPARED BY	DGM (TMD)
5. CHECKED BY	AGM (TMD)
6.RECHECKED BY	GM(MAINT)
7. REVIEWED BY	GM(O&M)
8. APPROVED BY	HOP
DATE OF ISSUE	
INDEX LIST	ENCLOSED
DISTRIBUTION LIST	ENCLOSED

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NTPC	LMI	SIMHADRI	
TITLE: Opration and maintenace practices of HVAC systems.			
REF NO. LMI/OGN/OPS/MECH/044:REV 02, DATE: FEB'2022		PAGE: 2 OF 7	
2.0 INTRODUCTION			
HVAC system is very important in power industry. Maintaining the required temperature and humidity in the control room is essential in avoiding electronic card failures and subsequent unit trippings. This calls for proper maintenance of HVAC system. Doing so,will help protect our building occupants from unnecessary exposure to IAQ (Indoor Air Quality) issues and further help protect us from potential litigation. There are some other very good reasons for proper maintenance of our HVAC system. These help in achieving the following. • Lower utility costs. • Increase in the service life of the HVAC equipment (reduce replacement costs). • Greater confort for the building's occupants. • Reduce electronic card failure. • Lower corbon emmission.			
3.0 SCOPE			
The scope of this LMI is to specify the various O & M practices for the HVAC systems installed at NTPC-Simhadri at locations like ST-I & II UCR & ESP AC system, ST-I & II DM plant AC system, Admin buildings & Sankalp buildings AC system, Switchyard building & Sakuntalam AC systems.			
4.0 SUPERSEDED DOCUMENTS.			
LMI/OGN/OPS/MECH/044, REV: 01, DATE: JAN'2022.			
5.0 REFERENCE DOCUMENT:			
OGN/OPS/MECH/044, Rev.00, Date: January'2011			
6.0 OPERATION AND MAINTENANCE PRACTICES FOR HVAC SYSTEMS.			
To maintain properly HVAC system the following practices are to be adopted at NTPC, Simhadri.			
Sl.No.	Activity	Reason / Maintenaace practices	PM Schedule
6.1	<u>Air filters of AHU</u> • Cleaning • Replacement	Filters captures dust, floating organic matter. - Pre filters (Main plant / others) - Fine filters Filters are made to be replaced frequently depending on dirt load.	8 / 12 Weekly. Half yearly. Condition based.

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Sl.No	Activity	Reason / Maintenaace practices	PM Schedule
6.2	<u>Air Duct (Main plant & others)</u> - Duct inspection - Duct cleaning - Dampers - AHU Inspection	Survey of air leaks and corrective actions. Supply and return air duct cleaning Cleaning, adjustment and lubricating Impeller cleaning, Bearings inspection and lubricating, belts inspection adjusment/alignment	Annually Once in 5 years (But it can vary based on actual condition) Annually 8/ 12 Weekly
6.3	<u>Condenser / Evaporators / Insulation of Duct and water line (Main plant & others)</u>		
	-Evaporator/ AHU	Cooling coil cleaning	8/ 12 Weekly
	-Condenser	Condenser tube cleaning	Condition based.
	- Insulation	Inspection of chilled water pipe line and duct's insulation. Repairing if found damaged.	Once in TwoYears
6.4	<u>Cooling Tower (Main plant & others)</u> - Nozzles, fan blades - Gear box - Fills and drift eliminators - Overhaul	Inspect / clean nozzles and fan blades and replace damaged nozzles. Inspect gear box, Fan belt condition and replace gear box oil and belt. Clean / adjust and replace damaged fills Overhauling the CT including basin	8/ 12 Weekly Six monthly 8/ 12 Weekly Based on condition

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Sl.No	Activity	Reason / Maintenance practices	PM Schedule
6.5	<u>Compressor (Reciprocating & Screw Chillers)</u> - Oil replacement - Foundation - Overhaul	Crank case oil replacement Looseness of foundation bolts Compressor overhauling	Condition based. 8/ 12 Weekly Once in four years / Condition based
6.6	<u>Walk down checks</u>	Operation/ Maintenance person will record DBT /WBT/RH of all Air conditioning spaces. Refrigerant and Oil level / Pressure checking	Daily
6.7	<u>Vapor Absorption Machine</u> - Healthness checking and sample collection of Absorbant (LiBr) - Vacuum pump, Absorbent pump and Refrigerant pump. - Condenser and Absorber	Checking of Healthiness of VAM machine and concentration of Absorbant(LiBr) Checking of Healthiness of pumps. Cleaning of tubes	Quartely 8 Weekly Condition based
6.8	<u>Ventilation system</u> - Suction Filters - AWU Blower - Nozzles, eliminators, Sump - Ventilation Duct	Cleaning Cleaning and Brgs. Lubrication Cleaning Inspection of Duct and repair if any damages	8 Weekly 8 Weekly 8 Weekly Annually
RESPONSIBILITY: HOD (TMD)			

NTPC	LMI	SIMHADRI
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7.0 EXCEPTION REPORTING

Any exception to the instructions specified in the LMI are to be reported as per the following format to HOD (MM) once in three months.

Resp: HOD (TMD)

Description of Exception	Unit No.	Location of exception	Remarks

8.0 RESPONSIBILITIES:

The responsibility of carrying out different tasks in relation to Operation & Maintenance of HVAC System as detailed in the LMI is with HOD (TMD)

9.0 RECORD KEEPING

The records to be maintained under this LMI include

- Compliance record of the activities detailed under this LMI
- Exception reporting.

The above records are to be maintained by MM- Turbine Maintenance.

10.0 REVIEW

Head of the project (HOP) shall review this LMI note once in two years or as and when the reference document is revised.

[illegible]

[illegible]